

Purchase of new Power Take-Off (PTO) to repair actively used TH60 drill rig

REQUIREMENT:

The PTO is used to transfer the truck engine's power via a transfer case to run drilling components, eliminating the need for a separate deck engine. The condition of the current PTO is beyond repair, and market research has shown that parts to repair/refurbish the current PTO for this specific rig are no longer made. The original manufacturer, however, currently has an unused PTO in their inventory that will be suitable as a replacement part.

Part #: 873-0121-012; replaced by 873-0039-012

Original Manufacturer: Fabco

Location:

Research Drilling Program
3595 E. Patrick Lane
Suite 100/200
Las Vegas, NV 89120

STATEMENT OF WORK:

To achieve the objectives outlined, the USGS Research Drilling Program (RDP) requires services from a contractor to replace the PTO for an actively used and highly needed TH60 drill rig.

This procurement will allow for the purchase of a new PTO and repair of the needed TH60 drill rig.

PTO requirements:

NOTE: specifications and requirements listed below are based on information provided by the manufacturer at

https://www.fabcotransfercase.com/Fabco_TC_170_170pd_Service_Manual.pdf.

1. Type & Functional Design
 - Single-speed transfer case designed for full-time all-wheel drive applications.
 - Available with or without:
 - PTO output
 - Hydraulic pump output
 - Proportional differential (33/67 or 50/50).
 - Air-actuated differential lock for traction control.
2. Lubrication Requirements

On-Highway:

- SAE 50 engine oil above +10°F
- SAE 30 engine oil below +10°F
- SAE 90 mineral gear oil above +10°F
- SAE 80 mineral gear oil below +10°F
- MIL-L-2105B EP oil permitted but not preferred

Oil Capacity

- Approx. 6.5 quarts, fill to level of fill plug.

3. Physical Size, Dimensions, and Compatibility

Mounting & Interface Requirements

- Upper, intermediate, front output, rear output, and lower (PD) shaft assemblies; each uses press-fit bearings and fixed housing bores.
- Yoke/companion flange sizes vary by model but use 1-3/4-12, 1-1/2-12, or 1-1/4-12 retaining locknuts depending on shaft.
- Bolt patterns:
 - Most carriers use 6-bolt or 10-bolt attachment patterns (7/16-14 cap screws).
 - Differential case uses 12-bolt patterns.
- Proportional differential versions require matching internal differential case halves, gears, and indexing dowels.
- Air shift assemblies require alignment of shift-shaft bores relative to case orientation ("TOP" markings present).
- Bearing & Gear Fitment
- All major gears and bearing cones are press-fit, requiring heat-fit installation (up to 300°F).
- Shaft end-float specification for all shafts:
 - 0.003–0.008 in. (select-fit shims determine final compatibility).

Driveline Configuration Compatibility - Works with chassis requiring:

- Full-time AWD
- PTO output or hydraulic pump drive (optional)
- Neutral provision (optional)
- PD options (33/67 or 50/50 torque split)

4. Weights

- ~520 lbs

5. Torque Specifications (Key Fitment Data)

Upper Shaft

- Carrier to housing: 45 lb.-ft. (7/16-14)
- Rear cap to housing: 45 lb.-ft. (7/16-14)
- Declutch housing: 20–35 lb.-ft.
- PTO yoke/flange locknut: 500 lb.-ft. (1-1/2-12)
- Yoke/flange (main): 600 lb.-ft. (1-3/4-12)

Front Output Shaft

- Declutch housing: 20 lb.-ft.
- Carrier to housing: 35 lb.-ft.

- Yoke/flange locknut: 600 lb.-ft. (1-3/4-12)

Rear Output Shaft

- Carrier bolts: 45 lb.-ft. (7/16-14)
- Yoke/flange locknut: 600 lb.-ft. (1-3/4-12)
- Shift cylinder cap: 15 lb.-ft. (1/4-20)

Lower Shaft (PD models)

- Lockout housing to case: 20–35 lb.-ft.
- Rear carrier bolts: 35 lb.-ft.
- Front and rear yoke locknuts: 400–600 lb.-ft. depending on spline

6. Operating Requirements

- Front driveline must be disconnected during chassis dynamometer operation.
- Both drivelines must be removed before towing.
- Differential lock may be engaged when stationary or moving (if axles aren't spinning).

Repairs/Work:

This request includes the labor associated with the removal of the faulty PTO, procurement, and installation of the available part.